

Hold Out Test Data

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compare a baseline for Final testing and honest claims. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Final testing and honest claims. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded claim.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

6. Solve the given case

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

Is it still untouched?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 23 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Calculate a final report

Calculate the worked accuracy, precision and recall with denominators.

My reasoning and evidence

13. W2 Compare a baseline

Why does 75 percent versus 70 percent not settle deployment?

My reasoning and evidence

14. W3 Handle a revision

The team changes a threshold after reading test errors. How should it label later results?

My reasoning and evidence

15. W4 Apply independently

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded claim.

My reasoning and evidence

16. W5 Report a failure

One planned case returns no prediction. What must the report retain?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Hold Out Test Data

Open the final hold-out only after recording the frozen model choice.

Frozen plant classifier evaluates 12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Frozen final-test results

	predicted spotted	predicted healthy
actual spotted	3	1
actual healthy	2	6

Read a cell using both its row and column labels.

Hold Out Test Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record freeze

Record model version and choices before reading these final references.



2. Calculate metrics

Accuracy $9/12=75\%$; precision $3/5=60\%$; recall $3/4=75\%$.



3. Bound claim

Report these twelve held-out cases and their collection conditions; preserve all three error records.

CHANGE ONE THING

The team changes a threshold after inspecting the false negatives.

Explain the changed result using the matching lesson.

What is needed for an honest final claim about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Open the final hold-out only after recording the frozen model choice.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report these twelve held-out cases and their collection conditions; preserve all three error records.

1. Record freeze
2. Bound claim
3. Calculate metrics

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound claim Report these twelve held-out cases and their collection conditions; preserve all three error records. Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The team changes a threshold after inspecting the false negatives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This set is now development evidence for the revised model.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: This set is now development evidence for the revised model. Using final outcomes to change the model breaks the untouched hold-out role.

Repair a missing building block

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared training candidate set. Short practice: May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on training targets. Short practice: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

